SANDEEP RATH

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rathsandeep.github.io | Chapel Hill, NC | Hyderabad, India

PROFILE

Operations researcher and clinical AI expert with 10+ years building and deploying data-driven models at UCLA Ronald Reagan Medical Center, UNC Health, UPMC, and the VA, generating millions in documented annual savings. Published in top-tier Operations Research journals including *Operations Research*, *POM*, and *M&SOM*. INFORMS Innovative Analytics Award Finalist. Active advisor to healthtech firms and the Government and Industry bodies in India on Health AI.

CORE EXPERTISE

Clinical AI	EHR data, Health AI Implementation and Workflow Integration, clinical outcome modeling, clinical decision support, clinical workload forecasting
Health Systems	Workforce scheduling, managing care coordination, digital health adoption, medical inventory management
Methods	Statistical Learning, Deep Learning, Causal Inference, Simulation, Mixed-integer and Stochastic Optimization, Dynamic Programming
Technical	Python, R, Julia, Gurobi, SQL, scikit-learn, PyTorch, pandas, git

EDUCATION

UCLA Anderson School of Management *2016*
PhD, Management (Operations Management) — *Dissertation: Resource Planning Models for Healthcare Organizations*

Indian Institute of Technology, Kharagpur *2006*
M.Tech., Instrumentation Engineering —*Dissertation: CART for Natural Text to Speech for Indian Languages* |
B.Tech., Electrical Engineering | Minor, Electronics & Telecommunication

EXPERIENCE

Academic Positions

Indian School of Business 2025–present
Assistant Professor, Operations Management *Hyderabad, India*

- *Research*: Clinical AI, clinical operations, digital workflow optimization, healthcare workforce planning, AI adoption in clinical settings
- *Teaching*: Operations Management, Healthcare Operations, Managing Services in the Digital World. Taught executive programs on AI in Healthcare for pharmaceutical industry and public health leaders.
- *Team Lead, ISB–Govt of Telangana AI in Healthcare Partnership* — Evaluating AI applications in public health systems for Government of Telangana State.
  - Developed an AI evaluation framework for public health in India
  - Evaluated 6 AI solutions covering oncology and maternal health
  - One solution selected for deployment; currently leading the pilot evaluation planning

Kenan-Flagler Business School, UNC Chapel Hill 2016–2025
Assistant Professor *Chapel Hill, NC*

- Collaborated with 4 major US health systems; deployed optimization models generating several million dollars in annual savings
- Taught Operations Management and Healthcare & Service Operations (avg. rating: 4.69–4.75/5); reviewer for *OR*, *MS*, *M&SOM*, *POM*

Industry Positions

Schlumberger Ltd. 2007–2010
Manufacturing Engineer *Pune; Houston; Paris*

· Lean six sigma process improvement; global supply chain IT systems management

ITC Ltd. 2006–2007
Assistant Manager, Operations *Chennai*
· Stabilised new manufacturing process for a $5M annual-revenue product line

ADVISORY & CONSULTING

Scientific Advisor & Consultant — Health & Technology 2025–present
· **Altysys** (technology) — led design and development of a prototype AI-driven care management dashboard.
· **Anchor, NABH Working Group on Digital Health Adoption** — Leading a community of practice for driving adoption of digital health standards and best practices in India.

SELECTED PUBLICATIONS — CLINICAL AI & SURGICAL OPERATIONS

1. **Rath, S.**, Swaminathan, J., Coleman, C. Optimal Staffing and Treatment for Collaborative Care of Diabetes and Depression. *POM*, 2026.
2. **Rath, S.**, Rajaram, K., Hudson, M.E., Mahajan, A. Multilocation, Dynamic Staff Planning for a Healthcare System. *Operations Research*, 2026.
3. Celik, U., **Rath, S.**, Kesavan, S., Staats, B. Does Physician's Choice of When to Perform EHR Tasks Influence Total EHR Workload? *M&SOM*, 2024.
4. Deshpande, V., Mundru, N., **Rath, S.**, et al. Data-Driven Surgical Tray Optimization. *Operations Research*, 2024. *— INFORMS Innovative Analytics Award Finalist 2022.*
5. **Rath, S.**, Rajaram, K. Staff Planning for Hospitals with Implicit Cost Estimation. *POM*, 2022.
6. **Rath, S.**, Rajaram, K., Mahajan, A. Integrated Anesthesiologist and Room Scheduling for Surgeries. *Operations Research*, 2017.
7. Deo, S., Rajaram, K., **Rath, S.**, et al. HIV Screening, Testing, and Care at the VA. *Operations Research*, 2015.

Working papers: Right Basket, Right Coupon: Basket-Aware Promotion Optimization in Online Retail (*submitted, M&SOM*); Reducing Appointment Delays: The Impact of Standardized EHR Usage on Physician Timeliness. *In progress:* Surgical Block Auto-Release Planning (with UNC Health); Fair On-Call Staffing with Heterogeneous Capability. *Other:* Annals of Plastic Surgery (2021); Journal of Vascular Surgery (2021); book chapter, *Foundations and Trends in TIOM* (2025); white paper on NABH Digital Health Standards, ISB (2026).

HONORS

Bullard Faculty Research Impact Award, UNC (2024) · Finalist, INFORMS Innovative Analytics Award (2022)
· POMS CHOM Research-in-Practice Showcase (2022) · UCLA Dissertation Year Fellowship (2015–16)

MEDIA COVERAGE

- New Research Finds Data-Driven Staffing Model Delivers Major Cost Savings for Healthcare Systems — *INFORMS News*, 2025
- A Major Medical Center Gets a Cheaper, Fairer Way to Assign Doctors — *UCLA Anderson Review*, 2025
- New Research Finds Data-Driven Staffing Model Delivers Major Cost Savings for Health Care Systems — *Medical Express*, 2025
- Data-Based Physician Staffing Stands to Save Millions of Dollars for Multi-Hospital Health Systems — *HealthExec*, 2025
- Collaborating to Cut Hospital Costs — *UNC Kenan-Flagler News*, 2022
- Surgeons Diagnose a Problem and Find a Business Solution — *UNC Kenan-Flagler News*, 2022
- A Scheduling Tool that Considers Employee Preferences — *UCLA Anderson Review*, November 2021